

Cramér–Rao Bound

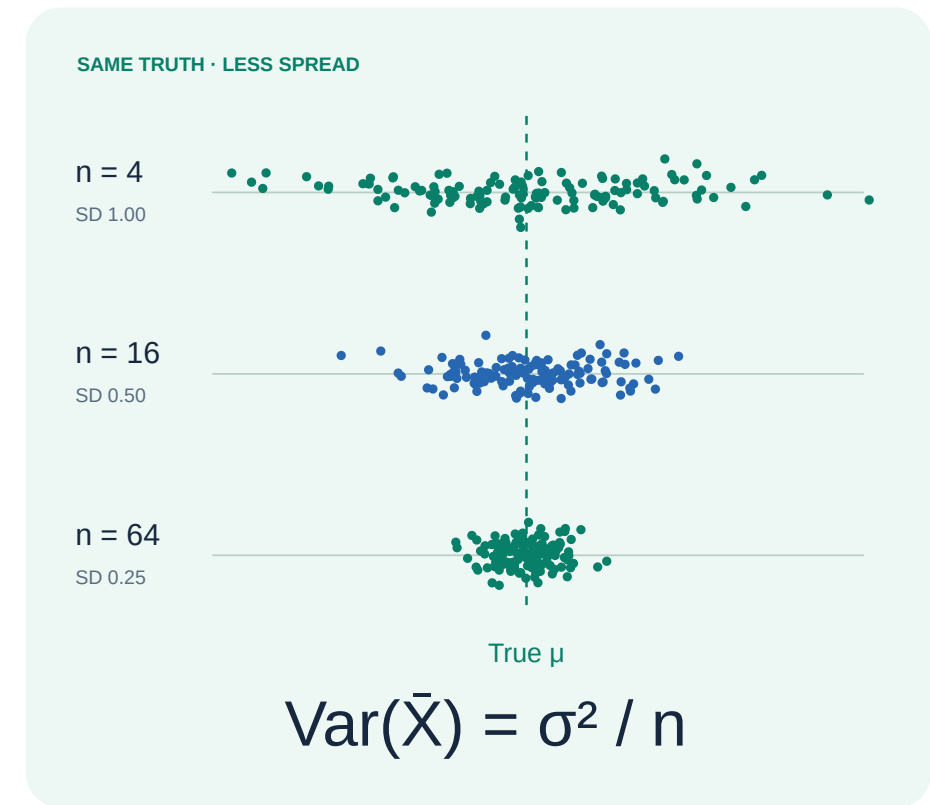
How precise could an estimate possibly be?

Precision has a floor.

Connect what measurements reveal to what estimators can achieve.

Fisher information · estimator bias
localization geometry

[Open the interactive guide](#) →



An estimate is one outcome. Precision is a distribution.

Keep the unknown parameter fixed. Repeat the entire measurement process.

01

A fixed world

The unknown θ stays fixed across trials.

02

New measurements

Each trial produces a fresh random dataset X .

03

A new estimate

Apply the same rule $T(X)$. Its spread is the variance.

$$T = T(X), \quad \mathbb{E}_{\theta}[T] = \theta$$

Unbiased means correct on average over repeated datasets.

How narrow can the cloud be?

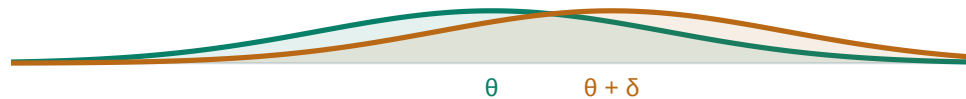
Precision describes repeated estimates. One close estimate does not establish precision.

Nearby worlds are hard to tell apart.

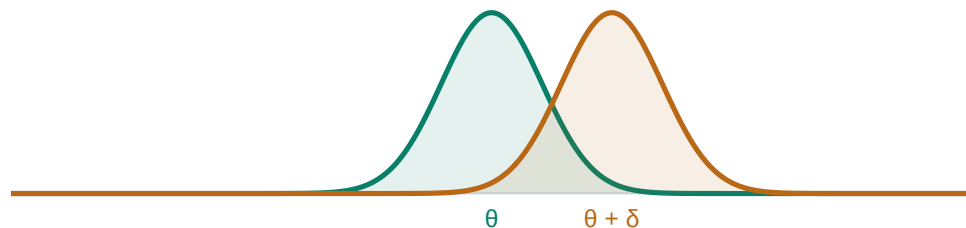
Information measures how sensitively the data distribution changes when the parameter moves.

Same separation · different noise

Noisier observations



Cleaner observations



Data densities; not posterior probabilities.

A change must leave a trace.

Almost indistinguishable data cannot support arbitrarily concentrated, locally unbiased estimates.

$$I_n(\theta) = \mathbb{E}_\theta \left[(\partial_\theta \log p(X; \theta))^2 \right]$$

More Fisher information permits a lower variance bound.

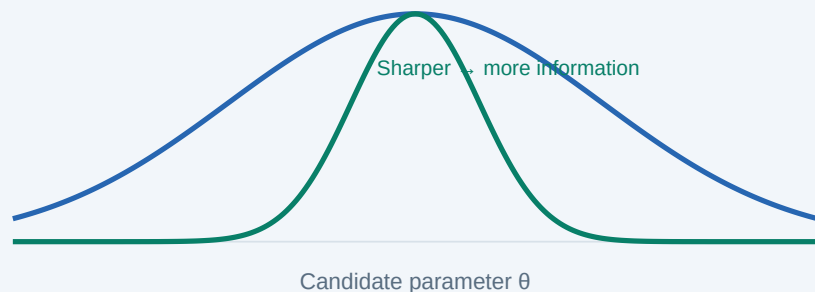
Likelihood → score → Fisher information.

The likelihood is a function of the candidate parameter, with the observed dataset held fixed.

1 · Likelihood

$$\ell(\theta; X) = \log p(X; \theta)$$

Likelihood / maximum



A likelihood is not a probability density over θ .

2 · Score

$$U_{\theta}(X) = \partial_{\theta} \ell(\theta; X)$$

Under regularity, the score has mean zero.

3 · Fisher information

$$I_n(\theta) = \mathbb{E}_{\theta}[U_{\theta}^2] = -\mathbb{E}_{\theta}[\ell''(\theta; X)]$$

Expected squared score = expected curvature, under the required interchange conditions.

The scalar Cramér–Rao Bound.

A floor for the variance of locally unbiased estimators in a regular model.

The variance floor

$$\text{Var}_{\theta}(T) \geq \frac{1}{I_n(\theta)}$$

$$\text{SD}_{\theta}(T) \geq \frac{1}{\sqrt{I_n(\theta)}}$$

Variance: squared units · SD: parameter units

The theorem has a scope.

Unbiasedness

$E_{\theta}[T] = \theta$ in a neighborhood.

Regularity

Differentiate under the expectation; $E_{\theta}[U_{\theta}] = 0$.

Finite information

$0 < I_n(\theta) < \infty$ and finite estimator variance.

Known model

The likelihood describes the experiment.

One score identity. One Cauchy–Schwarz inequality.

An unbiased estimator must move, on average, at the same rate as the true parameter.

The score identity

1 Differentiate unbiasedness

$$\partial_{\theta} \mathbb{E}_{\theta}[T] = 1$$

2 Move the derivative inside

$$\mathbb{E}_{\theta}[TU_{\theta}] = 1$$

3 Use the zero-mean score

$$\text{Cov}_{\theta}(T, U_{\theta}) = 1$$

Cauchy–Schwarz

$$\text{Cov}(T, U_{\theta})^2 \leq \text{Var}(T) \text{Var}(U_{\theta})$$

$$1 \leq \text{Var}(T) I_n(\theta)$$

$$\boxed{\text{Var}(T) \geq 1/I_n(\theta)}$$

An unbiased estimator must respond to the score.

For a Gaussian mean, averaging is already optimal.

Assume independent measurements $X_i \sim N(\mu, \sigma^2)$, with σ known.

Compute the information

$$\ell(\mu) = C - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

$$U_\mu = \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)$$

$$I_n(\mu) = \frac{n}{\sigma^2}$$

Compare with the sample mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \mathbb{E}[\bar{X}] = \mu$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} = \text{CRB}$$

Exact equality for every $n \geq 1$.

What will happen when you collect more measurements?

In the next slide, every histogram bar comes from genuinely simulated datasets.

Predict → change → explain

16 → 64 samples

What happens to variance and SD?

Double the noise σ

Which quantity becomes four times larger?

Use only the first observation

Does collecting unused data help?

Reference experiment

16

samples / trial

2

noise σ

1

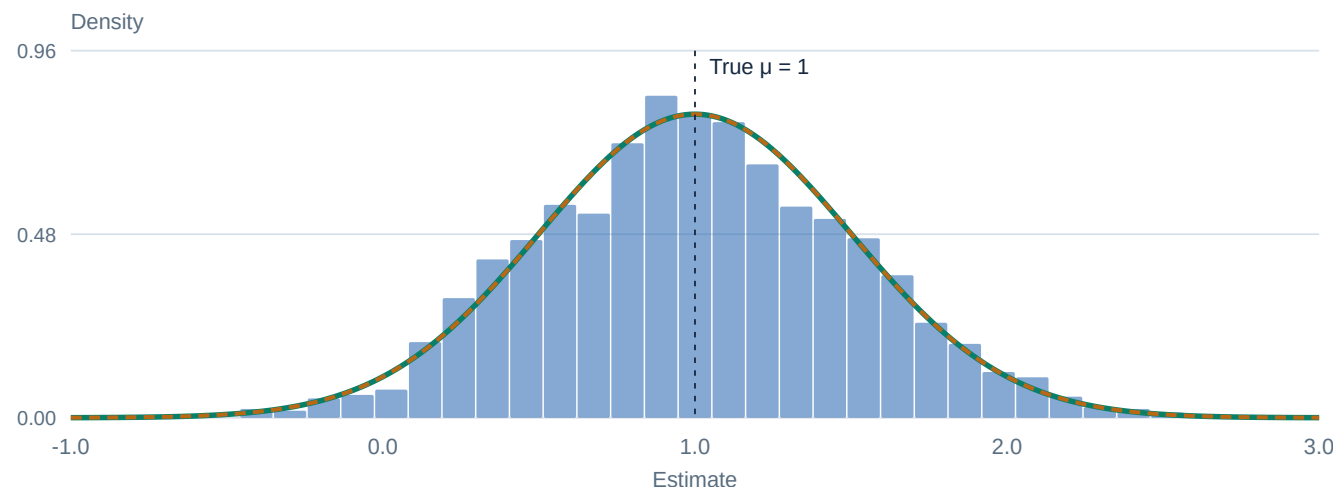
true μ

$$I_n = 4, \quad \text{CRB} = 0.25, \quad \text{SD}_{\min} = 0.5$$

Truth stays fixed. Each trial uses fresh observations.

Can you reach the precision floor?

Change the experiment, repeat it, and compare empirical spread with the exact bound.

**0.2570**

empirical variance

0.2500

exact variance

0.2500

unbiased CRB

Change the experiment

Independent Gaussian datasets.

$n = 16 \cdot \sigma = 2 \cdot \mu = 1$

2,000 trials · seed 42

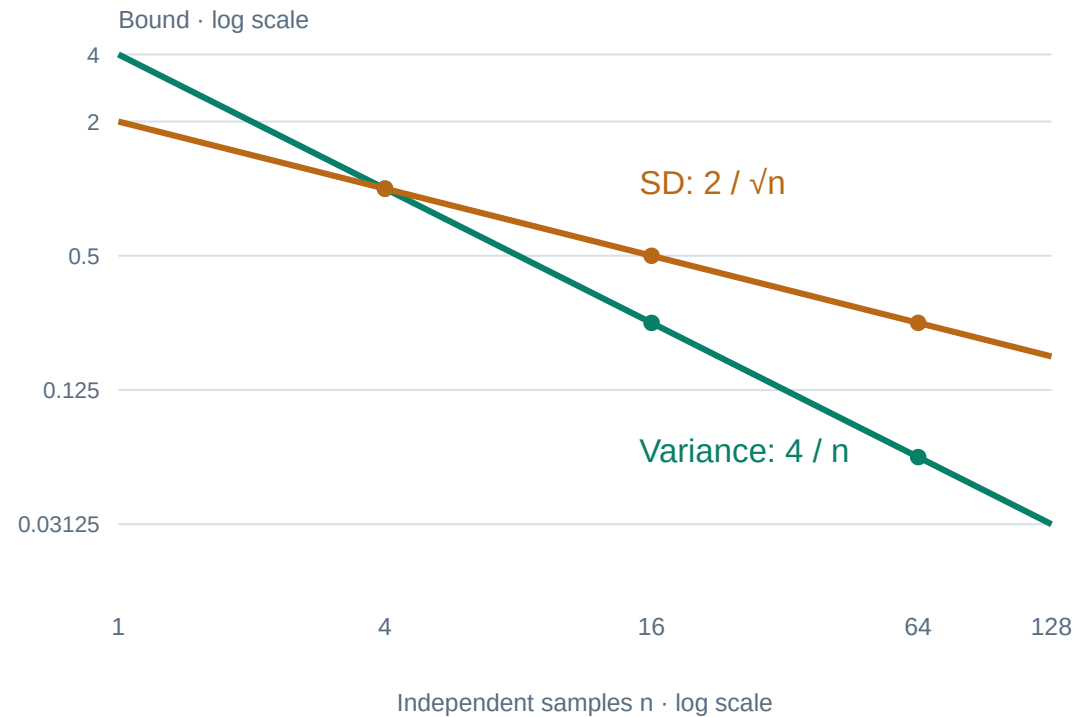
Increase n , change noise, or discard all but one observation. Compare the estimator with the same full-data bound.

Interactive controls in the [HTML slides](#).

A finite simulated variance can fall below the CRB through Monte Carlo fluctuation.

Four times the data. Half the standard deviation.

Information adds for independent measurements; variance and standard deviation scale differently.



$$I_n = nI_1, \quad \text{CRB} \propto n^{-1}, \quad \text{SD}_{\min} \propto n^{-1/2}$$

n	Variance bound	SD bound
4	1.0000	1.00
16	0.2500	0.50
64	0.0625	0.25

Correlated or non-identical data require the information of the joint likelihood.

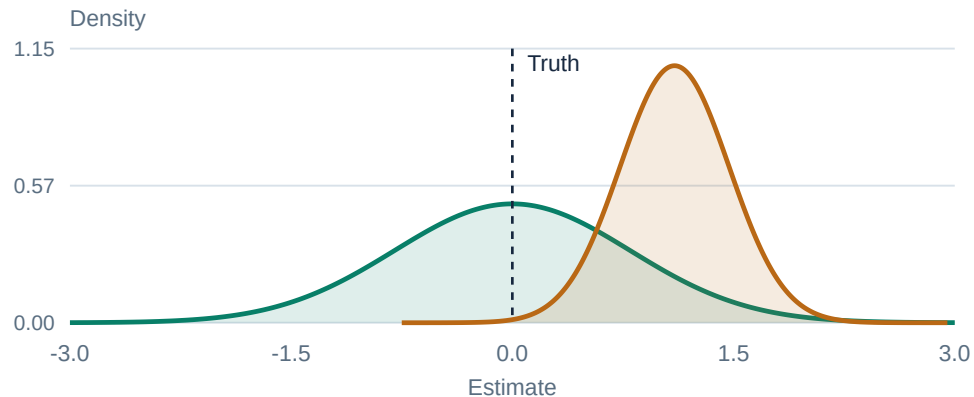
A narrow cloud can be centered in the wrong place.

The unbiased CRB is not a universal lower bound on mean squared error.

Bias–variance decomposition

$$b(\theta) = \mathbb{E}_{\theta}[T] - \theta$$

$$\text{MSE}_{\theta}(T) = \text{Var}_{\theta}(T) + b(\theta)^2$$



The correct biased extension

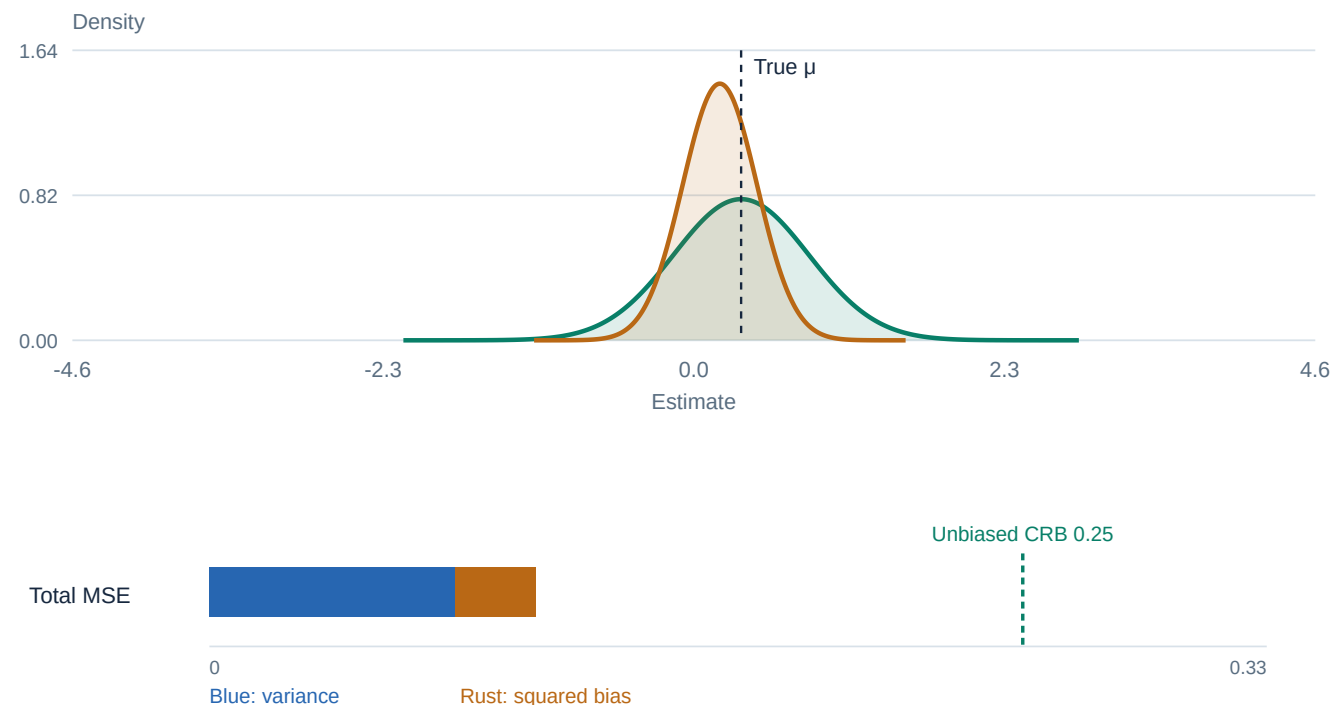
$$\text{Var}_{\theta}(T) \geq \frac{[1 + b'(\theta)]^2}{I_n(\theta)}$$

$$\text{MSE}_{\theta}(T) \geq \frac{[1 + b'(\theta)]^2}{I_n(\theta)} + b(\theta)^2$$

The derivative of the bias matters.

Can you “beat” the bound by shrinking toward zero?

Watch the variance fall, then move the true mean away from the shrinkage target.



Shrink toward zero

$$\bar{T}\alpha = \alpha \bar{X}$$

$$\alpha = 0.55 \cdot \mu = 0.35$$

$$\text{Variance} = 0.0756$$

$$\text{Squared bias} = 0.0248$$

$$\text{MSE} = 0.1004$$

Move μ away from zero.
Does the MSE advantage remain?

Interactive controls in the HTML slides.

The unbiased CRB is 0.25. A biased estimator can have a smaller pointwise MSE.

“Lower bound” does not mean “always achievable.”

Equality requires a very specific relationship between the estimator and the score.

Equality in Cauchy–Schwarz

$$T(X) - \theta = \frac{U_{\theta}(X)}{I_n(\theta)} \quad \text{almost surely}$$

Gaussian mean: $\frac{U_{\mu}}{I_n} = \bar{X} - \mu$

The same statistic must work throughout the parameter range.

Keep three claims distinct.

Unbiased

Expectation equals the target.

Efficient

Variance attains the CRB.

Maximum likelihood

Maximizes likelihood; finite-sample bias is possible.

Regular MLEs may attain the bound asymptotically.

Information has directions.

A vector parameter has a covariance matrix—not a single uncertainty number.

The matrix CRB

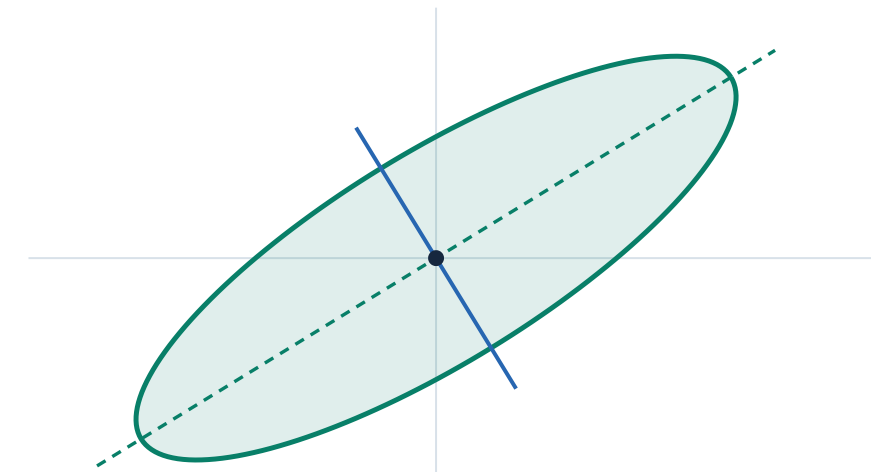
$$\mathbf{J}(\boldsymbol{\theta}) = \mathbb{E}[\mathbf{u}\mathbf{u}^T]$$

$$\text{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{J}(\boldsymbol{\theta})^{-1}$$

$$\mathbf{q}^T \text{Cov}(\hat{\boldsymbol{\theta}}) \mathbf{q} \geq \mathbf{q}^T \mathbf{J}^{-1} \mathbf{q}$$

Nonsingular $\mathbf{J}$; regular model; locally unbiased estimator.

Covariance-bound ellipse



Weak information → long error axis

Strong information → short error axis

Larger information eigenvalue → smaller directional variance bound.

One range measurement constrains one local direction.

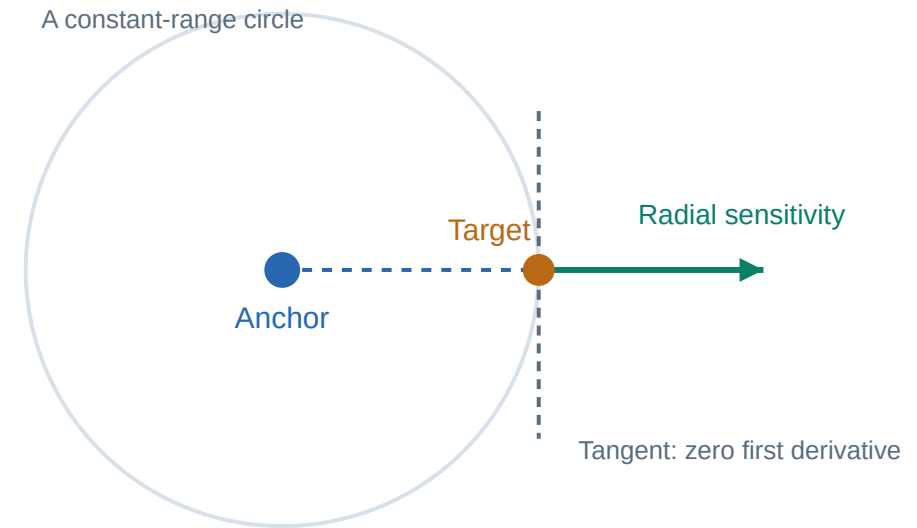
A simplified 2D ranging model: known anchor positions, independent Gaussian range noise.

Independent Gaussian ranges

$$z_i = \|\mathbf{p} - \mathbf{a}_i\| + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_i^2)$$

$$\mathbf{u}_i = \frac{\mathbf{p} - \mathbf{a}_i}{\|\mathbf{p} - \mathbf{a}_i\|}$$

$$\mathbf{J}(\mathbf{p}) = \sum_i \frac{1}{\sigma_i^2} \mathbf{u}_i \mathbf{u}_i^\top$$



Each term is rank one. Anchors are known; noise variance is known and independent of position.

Same sensors. Different geometry. Different limits.

Four accurate ranges can still leave one position direction poorly constrained.

Predict the weak direction

Surround

Anchors point toward the target from different directions.

Cluster

Anchors share almost the same line of sight.

Collinear

Anchors and target lie on the same straight line.

Position error bound

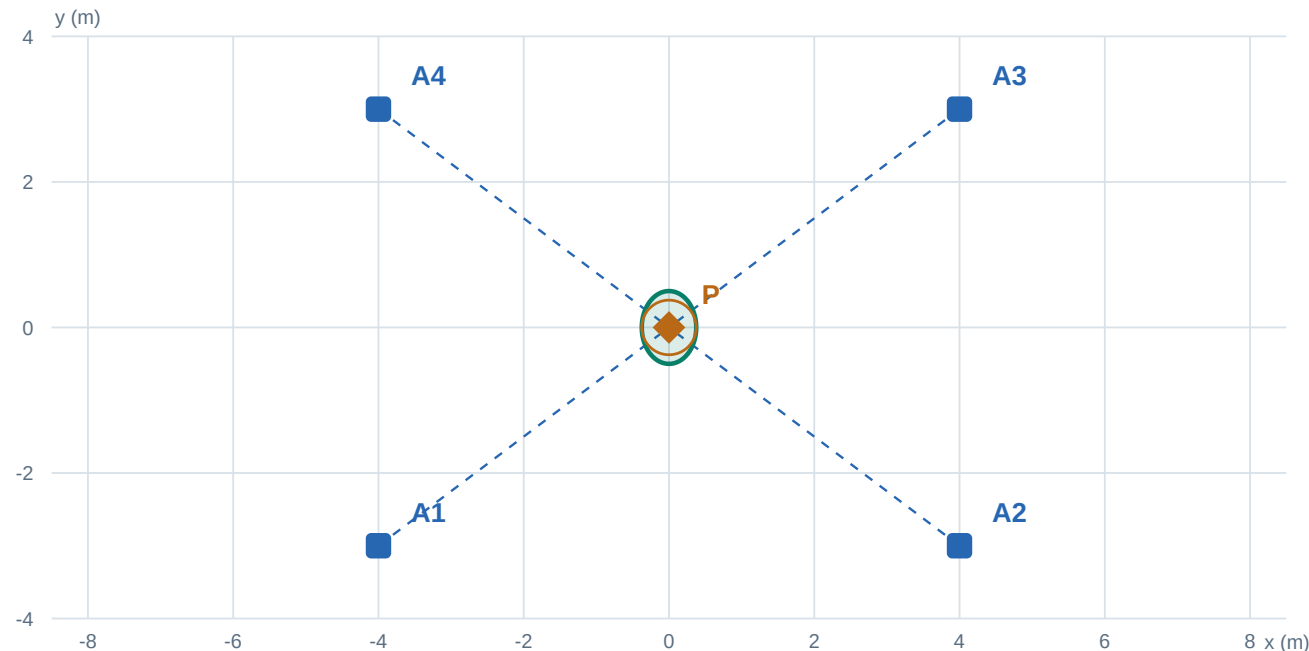
$$\text{PEB} = \sqrt{\text{tr}(\mathbf{J}^{-1})}$$

For an unbiased position estimator:
Euclidean RMSE $\geq$ PEB.

Finite local information does not rule out distant mirror solutions.

Move the sensors. Reshape the bound.

Drag any anchor or the target. The Fisher matrix and its inverse update immediately.



Default: $\sigma = 0.60$ m · PEB = 0.625 m

Reshape the information

Move an anchor or the target.
Compare Surround, Cluster, and
Collinear.

Then estimate an unknown common
range offset. Watch the equivalent
position information decrease.

Interactive controls in the HTML slides.

The ellipse is not a confidence region.

An unknown clock offset consumes position information.

The same measurements must now explain both the target position and a common range offset β .

Partition the information

$$\theta = \begin{bmatrix} \mathbf{p} \\ \beta \end{bmatrix}, \quad \mathbf{J} = \begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{b}^\top & c \end{bmatrix}$$

$$\mathbf{J}_e = \mathbf{A} - \mathbf{b} c^{-1} \mathbf{b}^\top$$

The position block of $\mathbf{J}^{-1}$ is $\mathbf{J}_e^{-1}$.

Some effects imitate each other.

An unknown offset can explain changes in range that would otherwise constrain position.

$$\mathbf{J}_e \preceq \mathbf{A}$$

No loss when $b = 0$. Clustered lines of sight can lose much more.

[Explore the geometry lab](#) →

A moving support breaks the familiar proof.

Not every model allows differentiation to pass through the expectation.

$X_i \sim \text{Uniform}(0, \theta)$

$$p(X; \theta) = \theta^{-n} \prod_{i=1}^n \mathbf{1}\{0 < X_i < \theta\}$$

$$U_\theta = -\frac{n}{\theta} \quad \text{inside the support}$$

$$\mathbb{E}_\theta[U_\theta] = -\frac{n}{\theta} \neq 0$$

The moving boundary invalidates the regular score identity.

Check the estimator directly.

$$T = \frac{n+1}{n} X_{(n)}, \quad \mathbb{E}_\theta[T] = \theta$$

$$\text{Var}_\theta(T) = \frac{\theta^2}{n(n+2)}$$

The variance follows from the distribution of the sample maximum.

Design the experiment, not only the estimator.

A new independent, correctly modeled measurement adds a positive-semidefinite information term.

Linear Gaussian measurements

$$\mathbf{z} = \mathbf{H}\boldsymbol{\theta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$$

$$\mathbf{J} = \mathbf{H}^T \mathbf{R}^{-1} \mathbf{H}$$

H sets sensitivity and geometry.
R sets noise and correlation.

Choose a design objective

Total variance

Minimize $\text{tr}(\mathbf{J}^{-1})$

Ellipsoid volume

Maximize $\log \det \mathbf{J}$

Weakest direction

Maximize $\lambda_{\min}(\mathbf{J})$

Local criteria depend on the parameterization and units.

Read a CRB result like a careful reviewer.

Before comparing an algorithm with a bound, make sure they solve the same statistical problem.

01 · Estimator class

Is the comparison valid?

Analyze bias before comparing a learned, constrained, or regularized estimator with an unbiased bound.

02 · Parameter vector

What was treated as known?

Include unknown offsets, calibration, orientation, and other nuisance parameters before inverting.

03 · Metric and model

Are the assumptions aligned?

Match variance, MSE, and RMSE with the correct bound. Model measurement correlations.

04 · Local versus global

What can the bound miss?

Inspect rank, conditioning, competing solutions, and low-information regimes.

Three ideas to keep.

More information lowers the bound. Bias changes the comparison. Geometry decides the weak directions.

01 · Sensitivity

Information belongs to the experiment.

$$I_n = \mathbb{E}[U_\theta^2]$$

Noise and sensing design change the limits before any estimator runs.

02 · Precision

Unbiased variance has a floor.

$$\text{Var}(T) \geq I_n^{-1}$$

An estimator within the theorem's class cannot beat the floor.

03 · Direction

The matrix tells you where.

$$\text{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{J}^{-1}$$

Complementary measurement directions improve the weakest constraints.

Continue from the bound to the bigger picture.

The equations and examples are teaching material; the sources below give the wider statistical context.

Primary references

[S1 · Stanford STATS 200, Lecture 15](#) →

Fisher information, scalar bound, and efficiency.

[S2 · Ly et al. \(2017\)](#) →

A Tutorial on Fisher Information.

[S3 · Shen & Win \(2010\)](#) →

Fundamental Limits of Wideband Localization, Part I.

[S4 · Stanford STATS 200, Lecture 14](#) →

Regular asymptotic normality of the MLE.

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